SR UNIVERSITY

School of Computer Science & Artificial Intelligence

Software Engineering — Mini Project Report

Team Details

S. No	Name of the Student	Roll Number	Email ID	Contribution
1	Manideep Reddy	2303A52183	2303a52183@sru.edu.in	Full Stack Development, Design & Documentation

Team Size: 1

Project Type: Web Application

1. Project Title

AI-Based Virtual Study Assistant

2. Problem Statement

Students often struggle with disorganized study schedules, lack of motivation, and inefficient information retrieval. The proposed AI-Based Virtual Study Assistant addresses these issues by providing an intelligent,interactive platform that helps students manage study time, clarify doubts, generate summaries, and track academic progress efficiently.

3. Objectives

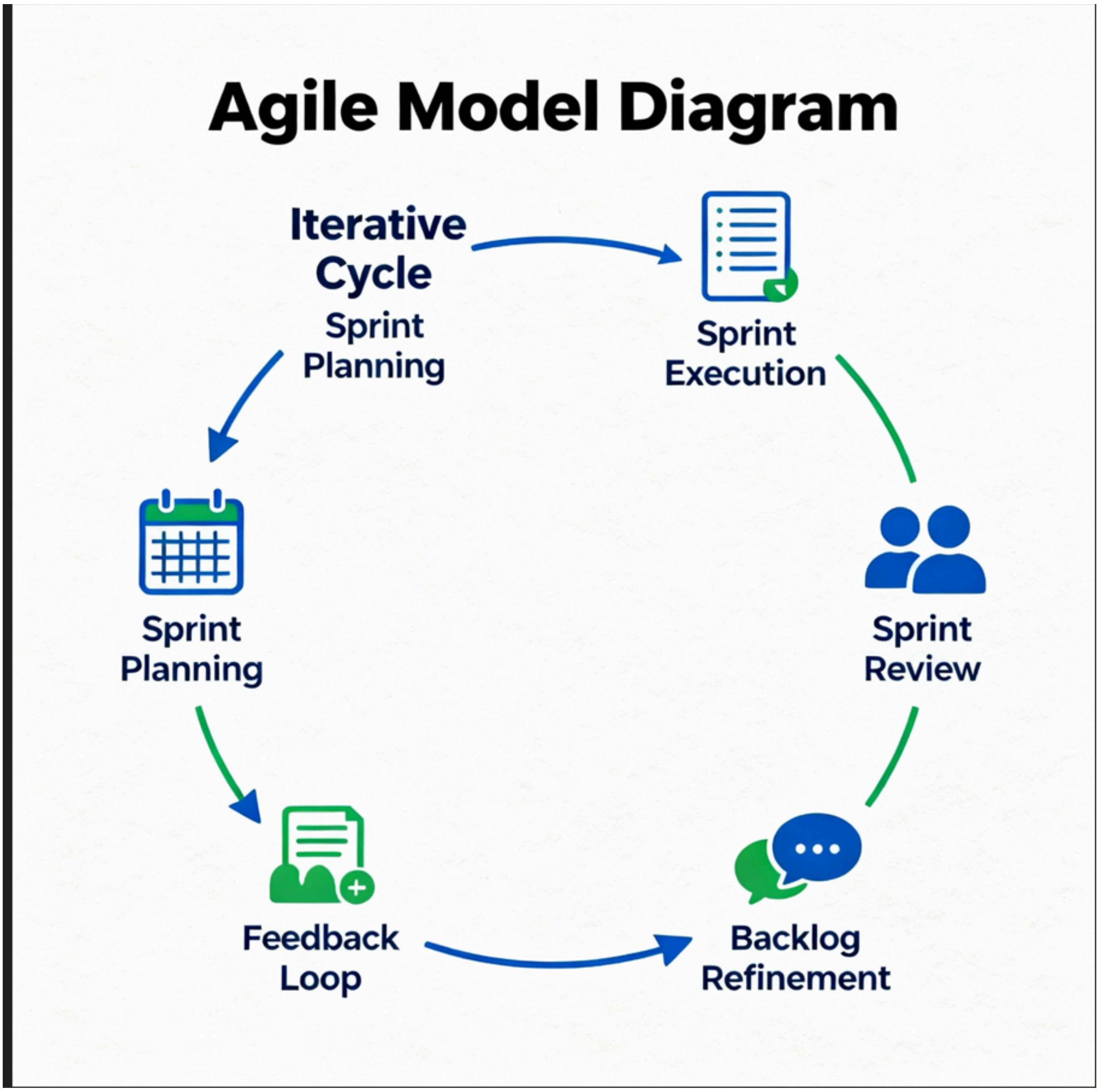
- To develop an AI chatbot that can assist students in study-related queries.
- To integrate reminder and scheduler functionality for study planning.
- To use NLP to summarize notes and generate answers.
- To build a responsive and user-friendly web interface.
- To enhance student productivity through AI-supported learning.

4. SDLC Model Adopted

Model: Agile Model

Justification:

The Agile model allows continuous iteration and feedback. It suits the dynamic nature of AI-based projects where training models and user feedback play crucial roles. Frequent improvements ensure adaptive learning and real-time refinement.


Agile Model Diagram
Iterative Cycle
Sprint Planning
Sprint Execution
Sprint Planning
Sprint Review
Feedback Loop
Backlog Refinement

5. Requirements Engineering

a. Functional Requirements:

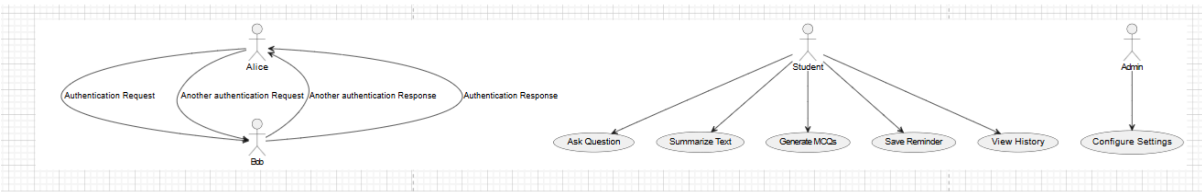
- User registration and login.
- Text and voice-based query interaction.
- Schedule planner and reminders.
- AI summarizer for uploaded notes.
- Performance analytics dashboard.

b. Non-Functional Requirements:

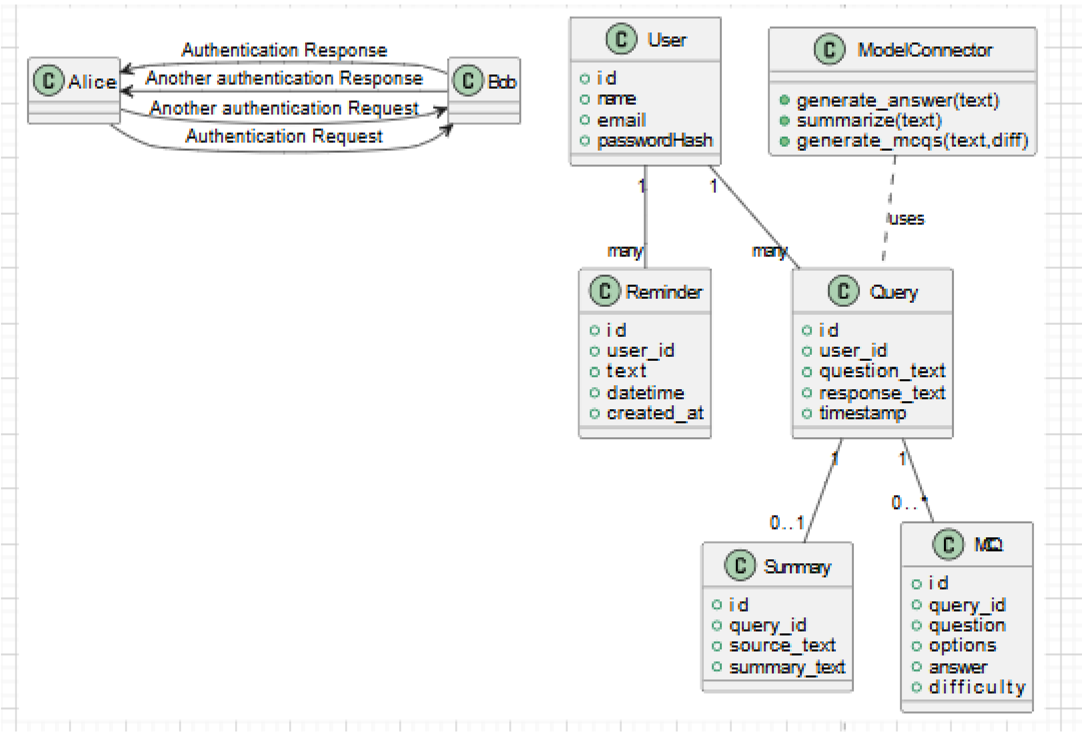
- The system should respond within 3 seconds for most queries.
- User-friendly and accessible via web browser and mobile.
- Secure authentication and data protection.
- Scalable architecture using cloud deployment.

6. System Analysis and UML Diagrams

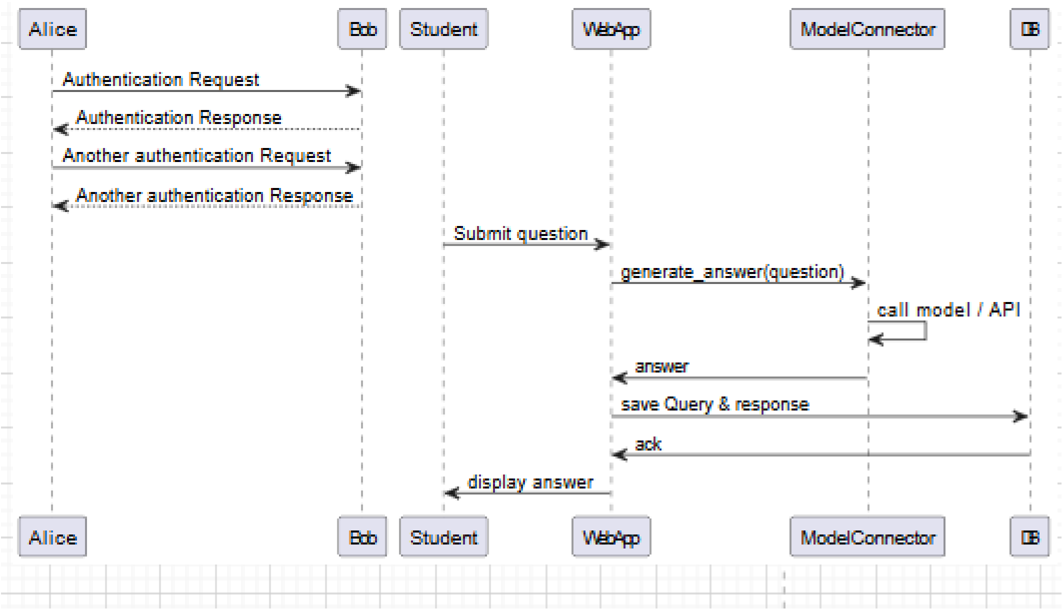
1. Use Case Diagram



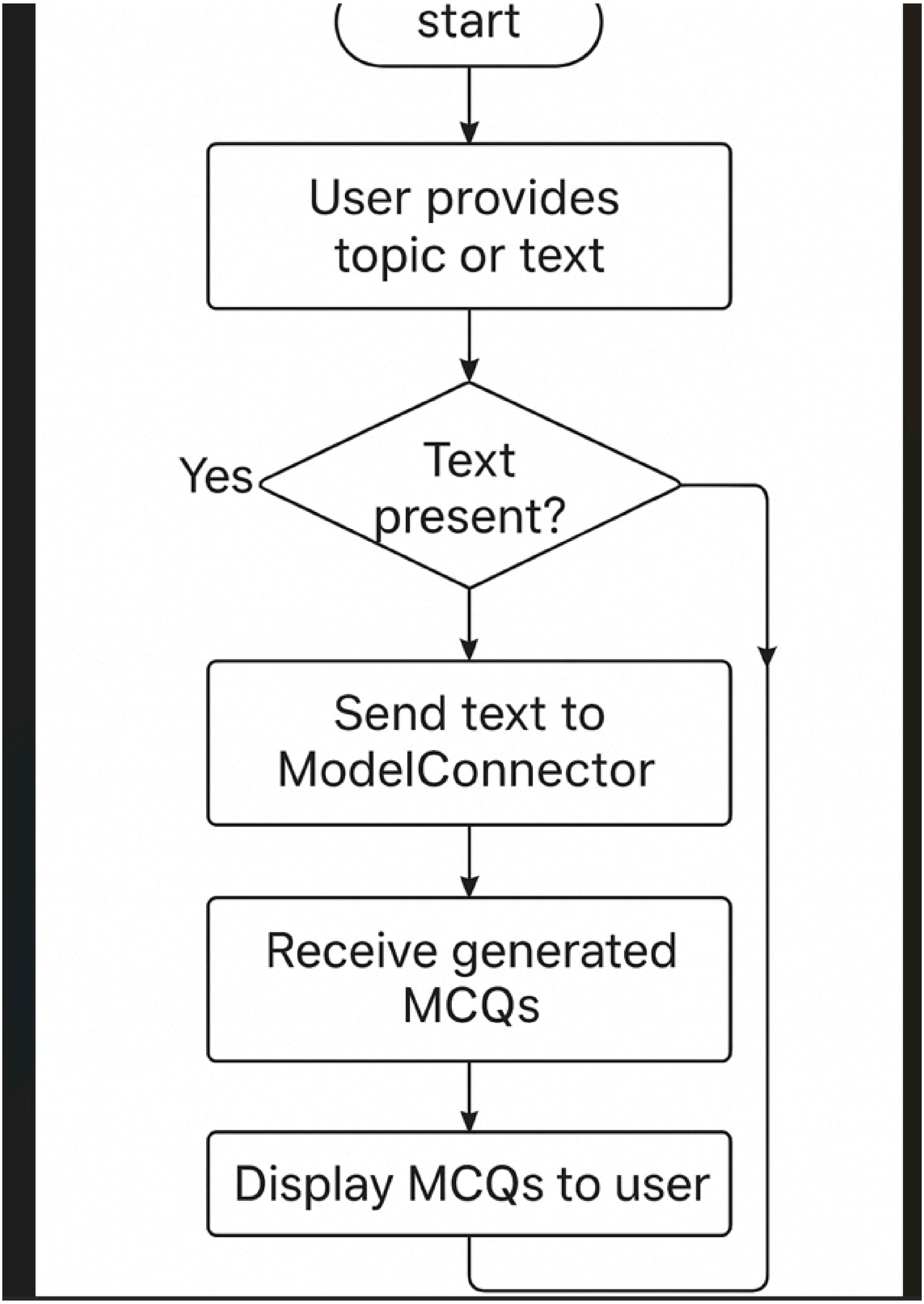
2. Class Diagram



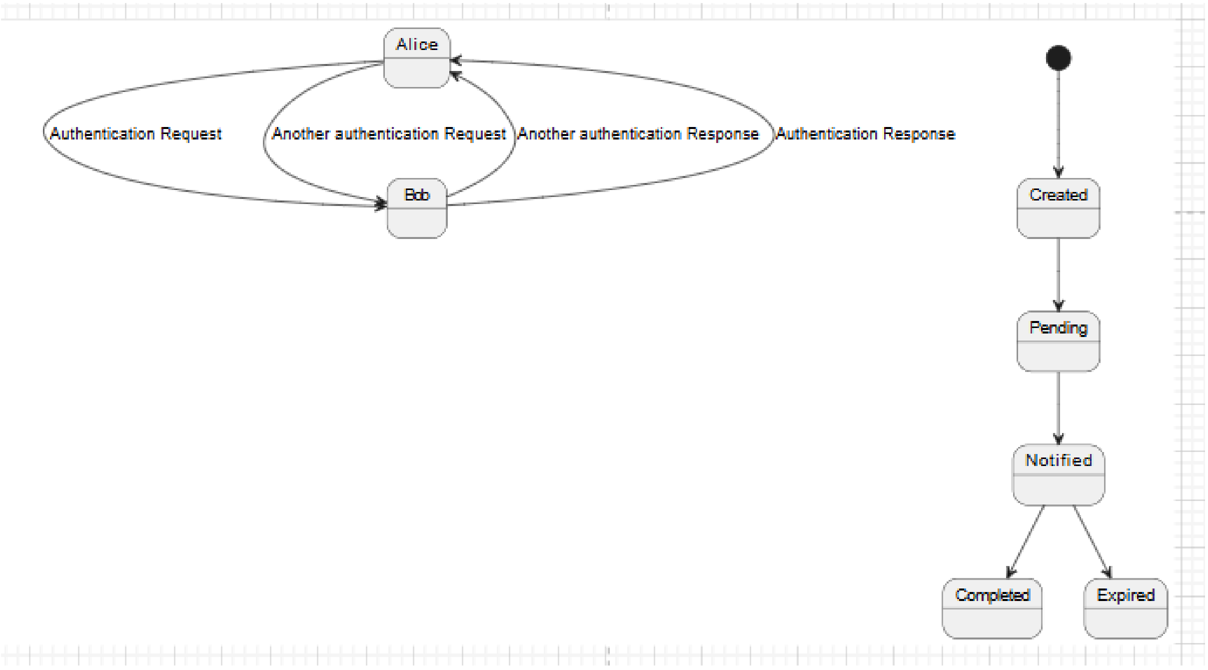
3. Sequence Diagram



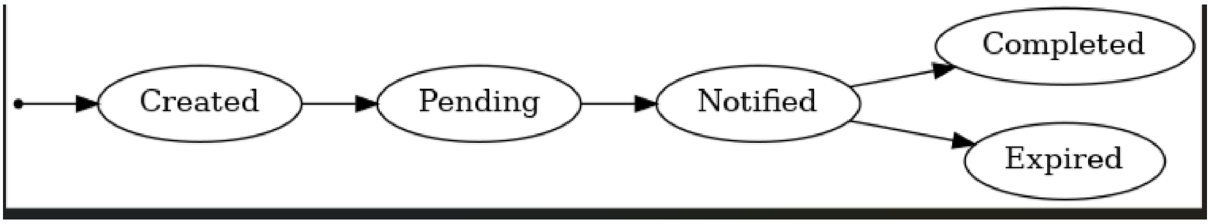
4. Activity Diagram



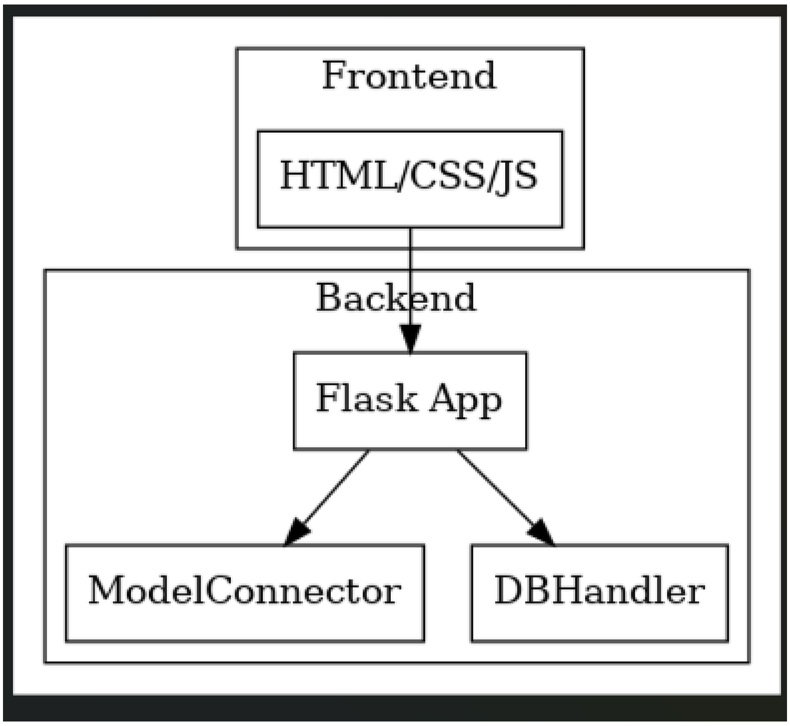
5. State Machine Diagram

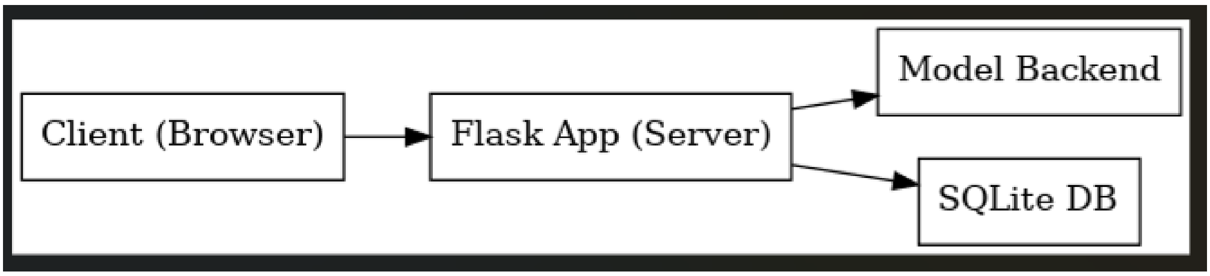


6. Deployment Diagram



7. Component Diagram





7. Design Principles Applied

- **Abstraction:** Chatbot hides NLP complexity.
- **Decomposition:** Modular separation of chatbot, scheduler, and summarizer.
- **Modularity:** Each module can be developed independently.
- **Cohesion & Coupling:** High cohesion within modules, low coupling between services.
- **Information Hiding:** User data secured via backend encryption.

8. Software Architecture

Architecture Style: MVC (Model-View-Controller)

Explanation:

MVC pattern separates business logic, data handling, and display layers ensuring scalability and maintainability.

9. Implementation

Languages & Tools:

HTML, CSS, JavaScript, Python (Flask), OpenAI API, MySQL, GitHub, Render/AWS deployment.

Key Features:

- Chatbot powered by GPT-like NLP model.
- Smart reminder module.
- Note summarization feature.
- Dashboard for analytics.

10. Testing

Techniques Used: Unit Testing, Integration Testing, System Testing

Tool: Pytest

Example Test Case:

Input: "Summarize Chapter 1 Notes"

Expected Output: "Provides key points with 90% accuracy."

11. GitHub Repository Repository Link:

https://github.com/2303A52183/SESD-PROJECT.git

12. Results and Discussion

The implemented system successfully provides AI-assisted study help, efficient task management, and academic insights. It improves learning engagement and saves time.

13. Challenges and Future Enhancements Challenges:

->Integrating AI model responses quickly.

-> Managing data storage for personalized summaries.

Enhancements:

-> Add voice command features.

-> Integrate OCR for handwritten note understanding.

-> Develop mobile application version.

14. Conclusion

The AI-Based Virtual Study Assistant bridges the gap between human tutors and smart learning tools by making studying more interactive, personalized, and efficient using AI technologies.

15. References

- OpenAI Documentation
- Flask Framework Docs
- Draw.io UML resources
- IEEE papers on Intelligent Tutoring Systems